



A Comprehensive Review on Blockchain-based Systems for Groundwater Conservation and Wastewater Management

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Abstract

Groundwater conservation and wastewater management soil depletion, water pollution, and poor resource management. These problems underscore the need for innovative practices that leverage blockchain technology to enhance sustainability, maintain data integrity, and optimize resource utilization in wastewater management. This review systematically analyzes recent advances (2019–2025) in blockchain technology, Machine Learning (ML), and Deep Learning (DL) models applied to sustainable water resource management. The objectives are to evaluate technological integrations, quantify performance improvements, and identify research gaps and future directions. Methodologies include a comprehensive review of 97 peer-reviewed articles focused on the integration of technology in water resource management. The review highlights benefits such as improved data validation, resource optimization, and transparency. Crucial applications include anomaly detection, groundwater recharge estimation, and regulatory compliance through blockchain-enabled secure data management. Technological contributions from smart sensors, internet of things-based real-time monitoring, Artificial Intelligence (AI)/ (ML) forecasting models, and advanced analytics collectively support more resilient and sustainable wastewater management practices. Key quantitative findings include predictive accuracy improvements up to 86%, wastewater treatment efficiency gains of 20%, and enhanced resource allocation through blockchain's decentralized frameworks. Despite these advances, challenges remain in data integration, scalability, and regulatory adoption, suggesting future research focus on integrating smart contracts, real-time monitoring systems, and internet of things-based governance frameworks to further enhance efficiency and compliance.

Keywords Blockchain · Groundwater Conservation · Wastewater Management · Deep Learning · Machine Learning

Introduction

With a permanent ledger of transactions and data exchanges, blockchain increases accountability and transparency in resource management (Patel et al., 2019). Its most notable features, such as decentralization, immutability, and increased security, make it a contemporary data management and protection tool across industries (Vangipuram et al., 2022). In the water sector, blockchain is used to address challenges in groundwater exploitation and

WasteWater Management (WWM) by ensuring that the use of water is correctly measured, the aquifer remains aware of such activity, and conservation efforts are verified (Poonia et al., 2021; Onifade et al., 2024). In WasteWater Treatment (WWT), its usage improves data integrity, compliance with set regulations, and operational efficiency (Hemamalini et al., 2024). One practical application involves the tracing of compliance through blockchain in municipal wastewater systems, which has, in turn, improved the transparency of audits and trust of the stakeholders (Onifade et al., 2024).

Alongside blockchain, ML and DL techniques are transforming groundwater preservation and WWT. Real-world use cases include attention-based ML models for early anomaly detection in WWT (Zhang et al., 2025) and genetic algorithm-based optimization of effluent quality (Ye et al., 2025). Long Short-Term Memory (LSTM)-based DL models perform precise nitrous oxide emission forecasting to assist in sustainable treatment practice (Seshan et al. 2025). ML tools show a relevant role in the

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conservation of groundwater since they signal groundwater level dynamics (Mahammad et al. 2023). Deep forest models based on feature selection supplemented with Bayesian optimization further improve predictions of groundwater recharge, which proves useful in having long-term conservation (Liu et al. 2024). Again, spatio-temporal approaches to predicting the vulnerability assessment are significantly improved using random forest methods for pre-emptive management of resources (Li et al. 2025).

Furthermore, classification-based ML models have been employed to assess wastewater quality, aiding treatment plant decision-making (Chen et al. 2025). while AI algorithms support predictive analysis and data-driven water governance strategies (Di Vaio et al., 2022). As blockchain technology matures, it is increasingly positioned to revolutionize WWM by delivering secure, transparent, and efficient frameworks (Balcerzak et al., 2022).

However, despite these advancements, significant research gaps remain. Current studies often lack integration of blockchain with real-time Internet of Things (IoT) monitoring and smart contract automation, limiting scalability and operational efficiency. Moreover, there is a scarcity of comprehensive frameworks that unify ML/DL predictive models with blockchain-enabled governance for water resource management. To address these gaps, this review sets the following SMART objectives:

- **Specific:** To evaluate the existing integration of blockchain, ML, and DL technologies for GroundWater Management (GWM) and WWM.
- **Measurable:** To analyse and synthesize findings from at least 90 peer-reviewed studies published between 2019 and 2025.
- **Achievable:** To identify key technological challenges and propose actionable future research directions.
- **Relevant:** To focus on enhancing sustainability, transparency, and operational efficiency in water resource management.
- **Time-bound:** To provide insights based on literature within the recent 6-year period, reflecting the most current technological advancements.

This structured approach aims to clarify the rationale behind this review and to guide future research toward practical, scalable solutions in sustainable WWM.

Contributions and Review Focus

This review thoroughly examines how blockchain, ML, and DL have been applied to groundwater conservation and WWT, analyzing 97 peer-reviewed studies from 2019 to 2025. The core contributions of this review are outlined below:

This review uniquely bridges the gap between blockchain and AI-based approaches by systematically evaluating their combined roles in ensuring transparency, data integrity, operational efficiency, and sustainability within GWM and WWT systems. Unlike previous studies that address these technologies in isolation, this review highlights their synergistic potential.

The review provides domain-specific insights by evaluating real-world applications such as the Blockchain Urban Water Management and Leakage Detection (BUWMD), Cost-Effective IoT-Blockchain Precision Agriculture System (CEIB-PAS), and Smart City Blockchain Water Management Architecture (SCBWMA). These case studies demonstrate the practical utility of blockchain in addressing issues like leakage detection, sustainability, and smart governance.

Employing a rigorous Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA)-based selection strategy, the review ensures methodological transparency and academic rigor. The inclusion-exclusion framework refines the scope to 97 high-quality studies, emphasizing methodological soundness, novelty, and practical relevance.

The review identifies critical gaps in existing literature, such as the limited scalability of blockchain frameworks in water systems, the high computational demands of DL models, and the lack of unified platforms for real-time data fusion. It outlines potential directions such as smart contract-driven automation and adaptive AI-blockchain ecosystems for next-generation water resource management.

Importance of Groundwater Conservation and Wastewater Management

Groundwater conservation and WWM are urgent global concerns, particularly as climate change, urban expansion, and agricultural intensification accelerate water stress. The United Nations World Water Development Report (UNWWDR) (2024) states that over 2.2 billion people lack safely managed drinking water, and 4.2 billion are without adequate sanitation. Furthermore, 80% of global wastewater is released untreated into the environment, leading to contamination of water bodies and loss of usable freshwater. Groundwater, which constitutes nearly 30% of the world's freshwater reserves, is rapidly depleting due to unsustainable extraction. International frameworks like Sustainable Development Goal 6 (SDG 6) emphasize the need for inclusive and sustainable WWM by 2030, urging innovations in recycling, monitoring, and regulation. These global trends underscore the necessity of data-driven, scalable, and secure systems to monitor, manage, and optimize water usage, particularly in regions facing acute water stress or lacking robust infrastructure.

Technological Imperatives and Emerging Solutions

The issues regarding water sustainability for innovative solutions, ensuring transparency, efficiency, and adaptability. Blockchain technology has emerged as a revolutionary solution, providing decentralized and immutable platforms where groundwater level monitoring, water rights management, and WWT compliance are tracked in real time by the parties involved. On the other front, ML and DL techniques are being employed to enhance intelligent decision-making through accurate forecasting, anomaly detection, and system optimization, among other things. Although these approaches face individual challenges, this review often addresses them independently, neglecting the synergy gained from combining them. This review fills in this gap by studying the intersection of blockchain, ML, and DL within the contexts of groundwater conservation and WWM. It describes how these technologies combine to improve predictive power, operational efficiencies, and regulatory compliance. Through a systematic review of 97 peer-reviewed articles from 2019 to 2025, the review identifies emerging frameworks, compares model performances, and stresses the real-life effects of integrated smart water systems.

Interconnected Challenges in Groundwater, Wastewater, and Soil Management

Groundwater conservation, WWM, and soil depletion are inherently interconnected due to their influence on the hydrological and nutrient cycles. Over extraction of groundwater lowers the water table, which impacts soil moisture, and increases salinization, compaction, and the depletion of cultivated soils' nutrients. At the same time, inadequate or untreated wastewater discharges lead to polluted surface and groundwater bodies with the addition of harmful chemicals and pathogens that decrease soil quality and disrupt microbial balance. By contrast, treated wastewater used for agricultural reuse replenishes groundwater levels and improves soil nutrient loading, and provides additional organic matter and nitrogen. Thus, an integrated approach that considers all three components is essential for achieving long-term agricultural productivity, ecological sustainability, and water security.

Process and Methodology

The methodology section follows established PRISMA guidelines to ensure a transparent and replicable selection process. It involves four key stages: literature identification, inclusion and exclusion criteria, screening and selection, and critical evaluation of the selected studies. The search and screening are independently conducted to minimize bias, with discrepancies resolved through consensus.

Literature identification

A comprehensive literature search is performed using only key academic databases, including IEEE Xplore, Springer, Elsevier, Wiley Online Library, Scopus, and Google Scholar. Domain-specific keywords are employed in the search strategy, such as “blockchain for WWM,” “ML in groundwater,” “DL WWT,” “blockchain-enabled WWT,” and “Groundwater Level Prediction (GLP).” The search is limited to studies that have been published between the years of 2019 and 2025 to stay concentrated on the most current technological developments. Broad or unrelated terms like “AI” and “automation” are excluded to avoid retrieving irrelevant studies from other engineering or financial fields. Search strings are adapted to each database's syntax, with Boolean operators and phrase matching applied. The search is conducted between January 10 and January 20, 2025.

Filtering and selection criteria

Inclusion criteria require studies to focus on groundwater or wastewater applications involving blockchain combined with ML or DL. Priority is given to research presenting real-world implementations, comparative evaluations, or quantitative performance results. Studies that lack practical focus, are purely theoretical, or do not address the integration of all three technologies are excluded. This ensures the review remains relevant and technically rigorous. Exclusion reasons include lack of empirical data, focus on unrelated domains, or absence of integrated technology approaches.

Screening process

The screening and selection process adheres to the PRISMA framework. Initially, 547 articles are identified through database searches. Title screening reduced the pool to 260, followed by abstract and conclusion reviews, bringing it down to 167, with 97 papers included after full-text assessment. Figure 1 depicts the PRISMA flow diagram outlining these phases. During screening, 287 records are excluded for irrelevance or duplication, 93 articles are removed at abstract screening due to insufficient focus, and 70 articles are excluded after full-text evaluation for lack of methodological rigor.

Critical evaluation

Each selected article undergoes critical analysis focusing on study objectives, methodologies, datasets, evaluation metrics, and the extent of blockchain integration with ML/DL. The extracted data are categorized into themes such as GLP, wastewater quality monitoring, blockchain-enabled

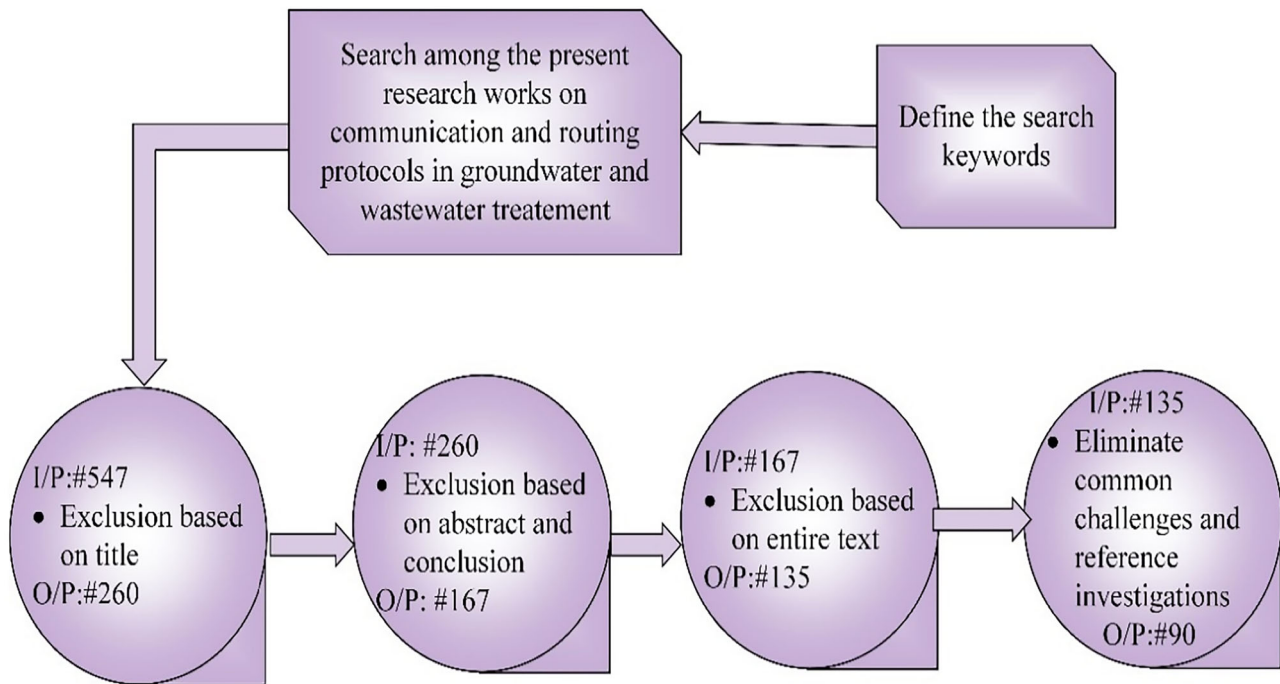


Fig. 1 Prisma process selection criteria

transparency, and smart contract automation. This systematic synthesis highlights current trends, practical applications, and research gaps. Quality appraisal is conducted using predefined criteria, including clarity of objectives, methodological robustness, dataset transparency, and evaluation comprehensiveness, ensuring consistent assessment standards across studies.

Outline of the Review

The review is designed to deliver a detailed summary of current technological developments in water resource management. Section 2 discusses the integration of blockchain, ML, and DL in GWM. Section 3 presents their applications in WWT. Section 4 summarizes key findings and technological trends from the reviewed literature. Finally, Section 5 offers conclusions and suggests future research directions.

Inclusion-exclusion Criteria and PRISMA Workflow

The PRISMA-based article selection process, covering identification, screening, and eligibility stages.

Figure 2 displays the complete methodological workflow employed in this review. The initial pool of 547 records is systematically refined through the defined steps to a final set of 97 peer-reviewed studies. This rigorous, methodical approach ensures comprehensive coverage and enhances the quality of the review's insights into blockchain and AI-

enabled WWM. Explicit counts and primary exclusion reasons are documented at each phase to support transparency and reproducibility of the selection process.

Groundwater Conservation-based Review

This review extends the workings of groundwater conservation and details some applications of blockchain technology in managing groundwater resources. Reviews ML techniques relating to Groundwater Quality Prediction (GQP) and uses remote sensing and mapping for GWM. Besides, discusses DL methods for forecasting groundwater levels using neural networks, provides an account of potential GWR estimation techniques, and focuses on DL applications for anomalous groundwater monitoring data identification.

Review Based on Blockchain Applications in GWM

There were recent advances in blockchain technology influencing WWM by showcasing various contemporary applications and frameworks intended for enhancing sustainability and efficiency. Dogo et al. (2019) underscored the role of AI-Based Water Management (AI-BWM), particularly their contribution to efficient real-time data acquisition and automation. This addressed the challenge of poor data availability and inefficient monitoring. Blockchain helped mitigate this by enabling tamper-proof

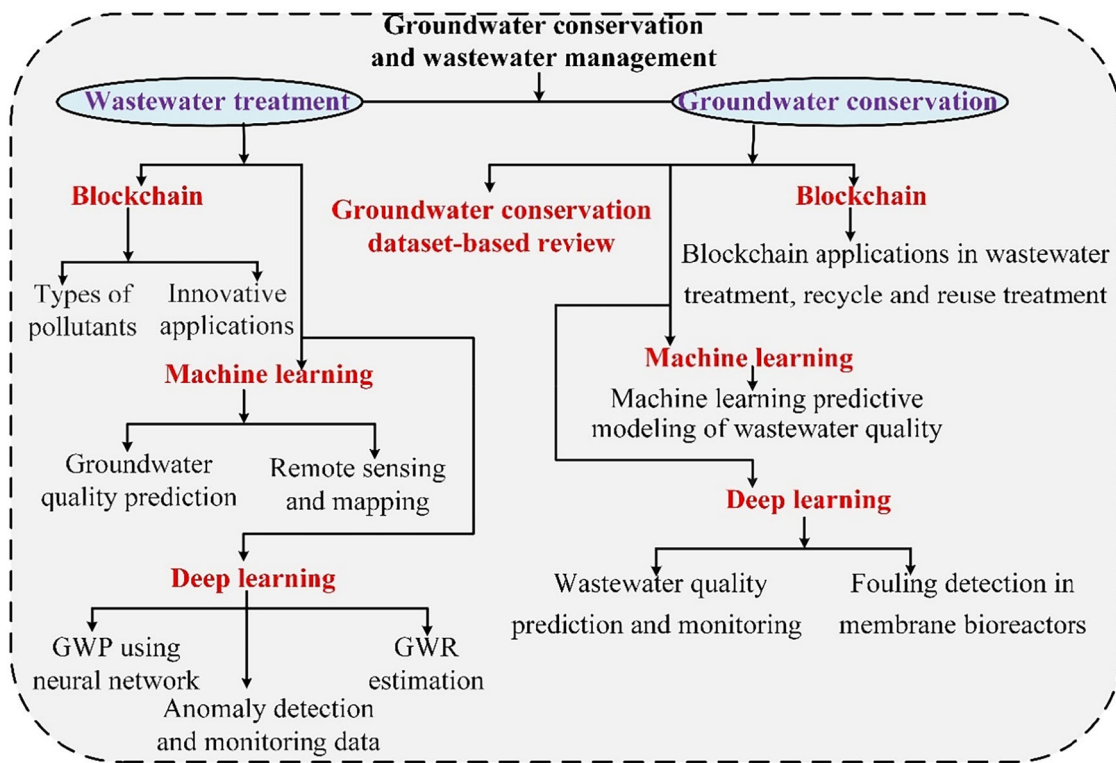


Fig. 2 Overall workflow of review based on wastewater treatment and groundwater conservation

real-time data acquisition and automation. Sriyono (2020) focused on building a Blockchain-Based Water Management System (BBWMS) to digitize the management of water, applying blockchain technology for the construction of sustainability. Making water use unsustainable, with a lack of transparency, is the model alleges. The blockchain technology provided traceability and secure data flow. Huge cost and scalability issues, however, were major considerations during implementation. Along similar lines, Sundaresan et al. (2021) suggested a blockchain framework for the SCBWMA, detailing how a blockchain enhances Smart Water Management (SWM) systems. It dealt primarily with data manipulation and the lack of transparency in smart city utilities. The blockchain gave data an immutable and auditable whereas a complex system architecture and privacy concerns gave rise to a new set of challenges. Chang et al. (2021) and Zeng et al. (2023) looked into agriculture with an IoT-blockchain nexus, taking SWM systems into account to enhance sustainability and efficiency in water usage. The problem addressed was inefficient water use in agriculture. Blockchain supported the secure sharing of sensor data and automation, enhancing sustainability. However, validation issues and sensor reliability presented difficulties. Pinchiera et al. (2021) outlined an economically feasible framework called the CEIB-PAS, which focuses on low-cost IoT devices as one of the reliable data sources of water

management systems. This addressed the problem of inaccessible data for small-scale farmers. Blockchain offered decentralized access and data security, though device costs and data integrity remained concerns. The comparative study of the blockchain application in governance management is found in Table 1.

Liu et al. (2021) developed the Digital Circular Economy for Water Sustainability (DCEWS) model, focusing on digitalization barriers in applying circular economy principles. They tackled the fragmented data systems and the lack of reuse models. Blockchain was used to enforce data traceability and lifecycle tracking. Yet, adaptation challenges and technical limitations were evident. Naqash et al. (2023) developed the BUWMD, which shows the functions of blockchain applications in addressing the problems of urban water management. This addressed issues like leakage detection and inefficient urban water use. Blockchain enabled secure, distributed tracking and data integrity, but struggled with high infrastructure demands and implementation costs.

ML-based Groundwater Quality Prediction

The reviewed literature in Table 2 highlights the rising application of ML methods for forecasting groundwater conditions across diverse geographies and environmental contexts. A notable trend is the consistent application of

Table 1 Comparison of blockchain applications in GWM

Refs	Focus	Type of pollution	Achieved results	Advantages	Limitations
(Dogo et al. 2019)	General water management	Various pollutants	Precision-85% Recall-78% Accuracy-80% F1 Score-81%	Enhanced management system, real-time monitoring.	Integration complexity, data security concerns.
(Sriyono 2020)	General water management	Sustainability issues	Precision- 80% Recall-75% Accuracy-77% F1 Score-77%	Improved sustainability and innovative management.	Faced implementation cost and scalability issues.
(Sundaresan et al. 2021)	Intelligent water management	Pollutants in smart cities	Precision-82% Recall-79% Accuracy-81% F1 Score-80%	Intelligent management and integration with smart cities.	Encountered complexity in architecture and data privacy challenges.
(Chang et al. 2021)	Agricultural water management	Agricultural runoff, contaminants	Precision-84% Recall-80% Accuracy-82% F1 Score-82%	Improved agricultural water management and real-time data.	Lacked extensive validation of the IoT and blockchain for SWM in agriculture.
(Pincheira et al. 2021)	Precision agriculture	Contaminants from agriculture	Precision-85% Recall-78% Accuracy-80% F1 Score-81%	Cost-effective, reliable data sources.	Incurred device costs and data integrity issues.
(Liu et al. 2021)	General water sustainability	Circular economy-related issues	Precision-86% Recall-82% Accuracy-84% F1 Score-84%	Address barriers to sustainability, circular economy focus.	Experienced implementation barriers and challenges in technological adaptation.
(Zeng et al. 2023)	Agricultural water management	Agricultural pollutants	Precision-83% Recall-80% Accuracy-81% F1 Score-81%	Enhanced SWM in agriculture.	Had integration issues and sensor reliability concerns.
(Naqash et al. 2023)	Urban water management	Leakage, inefficiencies	Precision-81% Recall-77% Accuracy-78% F1 Score-78%	Efficient management and leakage detection.	Struggled with urban infrastructure challenges and costs.

preprocessing and feature extraction techniques in most studies, which enhances model accuracy and robustness. Models that incorporate these steps, such as those developed by Pham et al. (2022), Das and Das (2024), and Cui et al. (2024), report high prediction accuracy, demonstrating the importance of data preparation in ML workflows. A key observation is that ensemble models and automated frameworks, such as those using closed-loop architectures or AutoML, show superior performance in handling complex hydrogeological data. These models improve generalization and reduce manual intervention, yet often come with increased computational complexity, limiting their practicality in resource-constrained settings. Despite promising results, several limitations persist. Many models are developed for highly specific contexts, e.g., coastal zones or drought-prone regions, which restricts their scalability and applicability to broader scenarios. Additionally, the lack of standardized performance metrics and datasets makes direct comparison difficult. While high accuracy is commonly reported, model interpretability and transparency are rarely

addressed, which is critical for adoption in environmental decision-making. These findings point to the need for future research to focus on (i) developing scalable and transferable ML models, (ii) enhancing interpretability through explainable AI techniques, and (iii) integrating real-time data from sensors and IoT systems to support dynamic groundwater quality management.

Table 2 served as a benchmark for ML-based GQP. Preprocessing, also includes steps of preparation and cleaning of data before injecting data to the ML model, like how to handle missing values, and data normalizing is done to enhance the overall performance of the model.

When marked with ✓, it indicates that pre-processing techniques were applied, which ✗ means that no pre-processing was utilized. By extracting key features from the original dataset, the model's performance in recognizing patterns and predicting outcomes is significantly improved. ✓ in this feature extraction column, signifies the use of feature extraction, while ✗ indicating it was not employed.

Table 2 Comparison of ML-based GQP

Refs	Focus	Pre-processing	Feature extraction	Advantages	Performances metrics	Limitations
(Vijay and Kamaraj 2019)	GQP	✗	✗	Utilized ML for precise prediction and was adaptable to various datasets.	Accuracy ~90%, RMSE ~ 2.1	Limited by data quality and required extensive preprocessing to ensure accurate predictions.
(El Bilali et al. 2021)	Groundwater quality for irrigation	✓	✓	It was effective for irrigation purposes and improved resource management.	$R^2 = 0.93$, RMSE = 1.45, MAE = 1.1	Specific to irrigation contexts and did not apply to other applications.
(Singha et al. 2021)	GQP	✓	✓	Employed efficient ML techniques and achieved accurate predictions.	Accuracy = 95.3%, $R^2 = 0.94$	Required significant computational resources and was dependent on the dataset, limiting its scalability.
(Pham et al. 2022)	GLP in drought areas	✓	✓	Predicted groundwater levels in drought-prone areas and was adaptable to different algorithms.	$R^2 = 0.92$, RMSE = 0.86 m, MAE = 0.51 m	Was limited to drought-prone areas, with model performance varying by region and not suitable for other environmental conditions.
(Das and Das 2024)	Coastal GQP	✓	✓	Combined weighted methods with ML and was suitable for coastal areas.	Accuracy = 94.6%, $R^2 = 0.91$	Focused on coastal areas, and the model did not apply to inland groundwater prediction.
(Singh et al. 2024)	Automated GLP	✓	✗	Automated the ML, reducing the need for manual tuning.	$R^2 = 0.95$, RMSE = 0.72 m	Required substantial computational power and did not handle all potential groundwater scenarios effectively.
(Cui et al. 2024)	GLP	✓	✓	Used ensemble ML for enhanced accuracy and adopted a multi-data approach.	$R^2 = 0.96$, RMSE = 0.61 m, MAE = 0.43 m	The model setup was complex and required high computational resources, limiting its accessibility for widespread use.

ML-based Remote Sensing and Groundwater Mapping

The review regarding the evolution and application of ML-based methods in remote sensing to address GPM issues, which brought about advancements in the predictive modeling dimension regarding data integration. Arabameri et al. (2019) applied ML and probabilistic models for possible groundwater mapping in their study, Damghan sedimentary plain, Iran, utilizing remotely measured data to identify groundwater recharge areas of suitability. Seo and Lee (2019) emphasized the advantages of remote sensing and GIS-based ML in accurate groundwater mapping. Mosavi et al. (2020) discussed groundwater salinity vulnerability employing Bayesian ML and ensemble methods. Sahour et al. (2022) utilized ML with multi-sensor remote sensing to identify groundwater in arid conditions. Sarkar et al. (2022) recently used ensemble ML methods for sustainable groundwater potential mapping. Nguyen et al. (2024) combine ML with remote sensing to produce spatial maps indicating groundwater availability across the Mekong Delta, Vietnam, signaling the importance of integrating multiple data sources to improve predictability. Ott et al. (2024) conducted a field-scale evaluation of GWM using the combined product of remotely sensed and climate data with an optimized pumping strategy. Table 3 compares ML-based remote sensing and groundwater mapping.

In Table 3, the pre-processing step was marked with a ✓ to indicate that it was an essential part of the methodology used in each referenced study. The ✓ mark denotes the application of preprocessing techniques. Preprocessing was the act of organizing raw data for adequate analysis, which includes cleaning, normalizing, and transforming based on the characteristics of the dataset. It guarantees

that the data employed in the models was coherent and of high quality, thereby supporting more accurate forecasting and outcomes.

DL-based GLP Using Neural Networks

This review elaborates further on the different uses of Artificial Neural Networks (ANN) and DL methods for groundwater quantity prediction and GWM. Bowes et al. (2019) utilized LSTMs in the context of flood forecasting for coastal cities and found that it was highly efficient for capturing flood behavior for groundwater table forecasting. Derbela and Nouiri (2020) trained ANN-based models in anticipation of future GLP and emphasized the great promise of intelligent algorithms in GWM. Di Nunno and Granata (2020) utilized the Nonlinear Autoregressive Exogenous input Neural Network (NAENN) in a localized predictive model for GLP in Southern Italy. Vu et al. (2021) used LSTM-DNN to lessen the effects of missing groundwater level data and improve prediction accuracy. Nourani et al. (2022) presented a DL-based conceptualization of how uncertainty quantification improves the reliability of predictive models for GLP. Gharehbaghi et al. (2022) used Gated Recurrent Unit Neural Networks (GRUNN) in their GLP model, which are sensitive to meteorological variability while incorporating it into predictive modeling. Chen et al. (2023) applied DL for GLP and added to the knowledge and applications of Adaptive Artificial Neural Network (AANN). Alabdulkreem et al. (2023) discussed the application of stacked LSTM combined with Deep Neural Network (DNN) for sustainable GWM, and again, the role of anatomy in resource management was emphasized. Seidu et al. (2023) examined how data partitioning affects GLP data prediction depending on the use of ANN models. The study was intended to allow optimal data

Table 3 ML-based remote sensing and groundwater mapping comparison

Refs	Developed methods	Pre-processing	Datasets used	Pros	Cons
(Arabameri et al. 2019)	Developed probabilistic and ML models for groundwater potentiality.	✓	Remote sensing data from Damghan sedimentary plain.	Improved GPM.	Limited to specific geographic areas.
(Lee et al. 2020)	Applied ML techniques for groundwater prediction management.	✓	GIS and remote sensing data sources	Enhanced potential mapping and geographic information system integration.	Limited details on optimization techniques.
(Mosavi et al. 2020)	Utilized classifier ensemble and Bayesian models to assess groundwater salinity susceptibility.	✓	Groundwater salinity data	Provided effective susceptibility mapping using ensemble models.	The complex model setup, not well.
(Sahour et al. 2022)	Employed multi-sensor remote sensing data and ML algorithms for identifying shallow groundwater.	✓	Multi-sensor remote sensing data	Facilitated improved identification in arid lands.	Limited to shallow groundwater in arid regions.
(Sarkar et al. 2022)	Implemented ensemble ML algorithms for groundwater prediction management.	✓	Groundwater potential data	Supported sustainable management with effective ensemble methods.	Limited details on optimization methods.
(Nguyen et al. 2024)	Using ML and remote sensing techniques to predict and manage groundwater resources.	✓	Remote sensing data from the Mekong Delta.	Utilized an integrated approach that enhanced potential mapping.	Region-specific, not applicable to other areas.
(Ott et al., 2024)	Used remote sensing and climate data for GWM.	✓	Remote sensing and climate data.	Improved GWM, field-scale applications.	High computational requirements.

Table 4 Comparison of GLP using neural networks

Refs	Pre-processing	Feature extraction	Optimizations	Datasets Used	Performance metrics	Limitations
(Bowes et al. 2019)	✓	✓	✓	Groundwater data from a coastal city.	Root Mean Square Error (RMSE): 0.95	Limited to coastal environments.
(Derbela and Nouiri 2020)	✓	✓	✓	Groundwater data from Mediterranean regions.	R ² : 0.89	Requires high-quality data for accuracy.
(Di Nunno and Granata 2020)	✓	✓	✓	Apulia region groundwater data.	RMSE: 1.2	Limited to specific regions.
(Vu et al. 2021)	✓	✓	✓	Groundwater data from Normandy, France.	RMSE: 0.83	Model performance depends on dataset quality.
(Nourani et al. 2022)	✓	✓	✓	Groundwater and climate data.	RMSE: 0.95	High computational cost.
(Gharehbaghi et al. 2022)	✓	✓	✓	Meteorological and groundwater data from Iran.	RMSE: 0.78	Requires extensive meteorological data.
(Chen et al. 2023)	✓	✓	✓	Groundwater data from Asia.	RMSE: 1.1 and R ² : 0.88	Computationally intensive.
(Alabdulkreem et al. 2023)	✓	✓	✓	Groundwater data from urban areas.	RMSE: 0.92	High computational resources were required.
(Seidu et al. 2023)	✓	✓	✓	Groundwater data from Ghana.	R ² : 0.90	Region-specific require large datasets.
(Abdi et al. 2024)	✓	✓	✓	Groundwater data from multiple regions.	RMSE: 0.88	Model complexity increases with data size.

use and to investigate data formatting as an area of future research. Abdi et al. (2024) implemented interpolation methods in DL models to improve the accuracy of GLP models. Table 4 provides a comprehensive analysis of GLP as it relates to any particular neural network applications.

In Table 4, ✓ the pre-processing column is used to denote that these methods were used. Feature extraction was the selection and transformation of pertinent features from raw data to enhance operations like pattern recognition and prediction.

A ✓ in the column for feature extraction indicates that feature extraction techniques were used. Optimization refers to procedures of fine-tuning model parameters to achieve good performance. A ✓ in the optimization column shows that optimization methods were employed.

DL-based Groundwater Recharge Estimation Review

The exploration center employs a range of different DL techniques to predict and estimate groundwater resources in

Table 5 Comparison of groundwater recharge estimation

Refs	Preprocessing	Feature extraction	Optimization	Limitations
(Huang et al. 2019)	Data normalization, handling missing data.	✓	✓	Had limited generalization in different hydrological conditions.
(Afzaal et al. 2019)	Standardization, outlier removal.	✓	✓	Required high computational resources due to complex models.
(Sihag et al. 2020)	Data cleaning, feature scaling.	✓	✓	Had limited applicability in areas with scarce data.
(Zhu et al. 2020)	Noise reduction, normalization.	✓	✓	Increased uncertainty due to hydrological model complexity.
(Elzain et al. 2021)	Standardization, interpolation of missing data.	✓	✓	Performance was region-specific and dataset-dependent.
(Huang et al. 2023)	Normalization, dimensionality reduction.	✓	✓	Required a large amount of spatial-temporal data.
(Al-Ruzouq et al. 2024)	Noise filtering, image preprocessing.	✓	✓	Depended on high-quality remote sensing data.
(Mirboluki et al. 2024)	Feature scaling, handling missing data.	✓	✓	Incurred a high computational cost due to complex DL models.

all of these studies. Huang et al. (2019) used GWR prediction encouraged by three separate methods: linear regression, multilayer perceptron networks, and direct DL methods, so that they can compare the three above. Afzaal et al. (2019) contributed to groundwater estimation using hydrology factors with an ANN model and through DL, creating new groundwater estimation structures and improving accuracy in estimating groundwater resources. Sihag et al. (2020) used the random forest method to estimate GWR rates and provided a consistent method for estimating recharge. Zhu et al. (2020) addressed the associated uncertainties with hydrological modeling regarding diffuse estimation of GWR, and the resulting estimates were more accurate.

Elzain et al. (2021) examined aquifer vulnerability indices utilizing a combination of DL and optimization algorithms for advancing aquifer vulnerability assessment. Huang et al. (2023) presented a top-down DL that aimed to predict GWR spatiotemporal behavior, stressing temporality and spatiality in aquifer recharge. Al-Ruzouq et al. (2024) integrated DL and remote sensing techniques to identify artificial GWR zones, which reflected an integrated recharge area identification method. Mirboluki et al. (2024) concentrated on the estimation of groundwater levels through advanced DL and soft computing methodologies that resulted in improved estimation methods. Table 5 provides a comparison of GWR estimation.

In Table 5, the ✓ in the feature extraction column signifies that feature extraction techniques were used. Optimization refers to techniques used to fine-tune the model parameters and improve the model's performance. A ✓ in the optimization column indicates that optimization methods were applied.

DL-based Anomaly Detection in Groundwater Monitoring Data

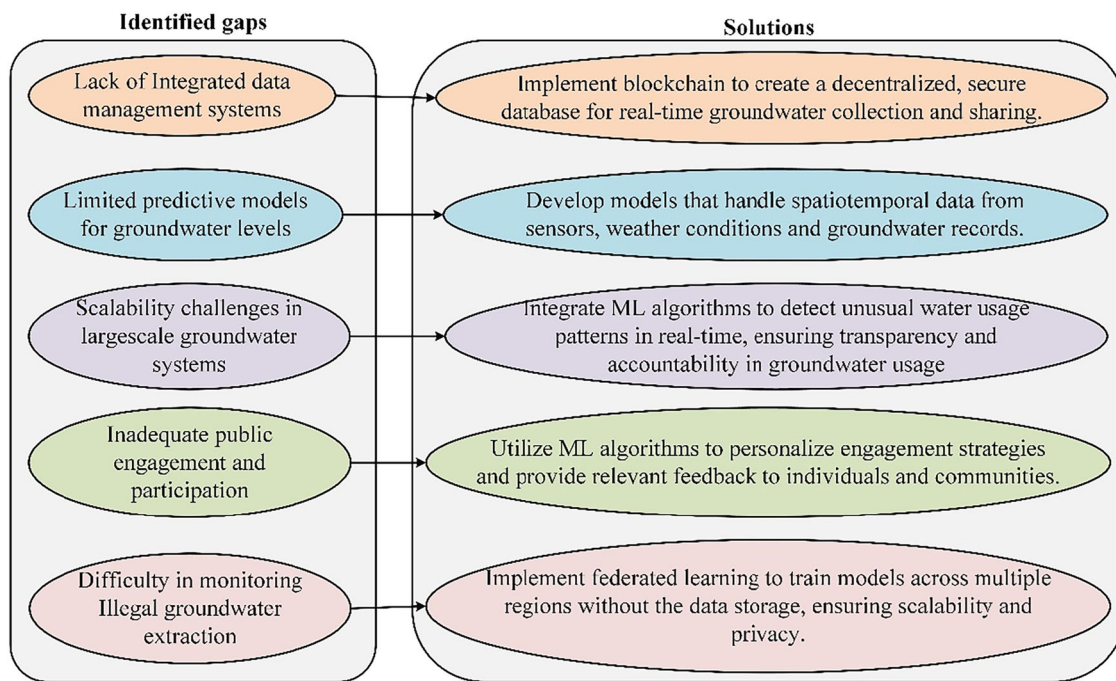
The literature emphasizes notable advancements in utilizing DL for monitoring water systems, with a focus on anomaly detection and predictive capabilities. Seo and Lee (2019) developed a spatio-temporal framework for tracking groundwater drought by using multi-satellite data with an ANN to effectively monitor groundwater levels. Similarly, Zhong et al. (2019) applied detecting irregularities in geological carbon storage through pressure readings using DL techniques. Besides, Miao and Hung (2020) improved flood forecasting and anomaly detection by implementing a DL model, a progress that Liu et al. (2020) also supported, and used data-driven methods to identify anomalies in surface water quality. El-Shafeiy et al. (2023) applied multivariate DL for real-time water quality monitoring. Sorkhabi et al. (2023) introduced a DL framework to track groundwater storage using satellite data. Prabu et al. (2024) combined IoT with k-NN-SVM for sustainable GWM. Rezaiezhadeh Roukerd and Rajabi (2024) enhanced anomaly detection in groundwater systems via LSTM-Autoencoder models. Table 6 compares these approaches in groundwater anomaly detection.

In Table 6, the ✓ in the feature extraction column signifies that feature extraction techniques were used. Optimization refers to techniques used to fine-tune the model parameters and improve the model's performance. A ✓ in the optimization column indicates that optimization methods were applied.

Figure 3 depicts the research gaps and solutions for groundwater conservation. “This review has highlighted

Table 6 Comparison of anomaly detection in groundwater monitoring data

Refs	Preprocessing	Feature extraction	Optimization	Limitations
(Seo and Lee 2019)	Satellite data cleaning, normalization.	✓	✓	Limited by satellite resolution and temporal gaps.
(Zhong et al. 2019)	Pressure data filtering, outlier removal.	✓	✓	Model performance varied with different geological formations.
(Miau and Hung 2020)	River flow data cleaning, time-series formatting.	✓	✓	High computational demands for real-time processing.
(Liu et al. 2020)	Water quality data normalization.	✓	✓	Limited by sensor noise and occasional false positives.
(El-Shafeiy et al. 2023)	Sensor data cleaning, noise reduction.	✓	✓	Limited scalability for large networks of sensors.
(Sorkhabi et al. 2023)	Satellite gravimetry data processing.	✓	✓	Dependency on satellite data accuracy.
(Prabu et al. 2024)	IoT data normalization, missing data imputation.	✓	✓	High complexity in developed model tuning.
(Rezaiezhadeh Roukerd and Rajabi 2024)	Time-series data normalization.	✓	✓	Model performance is reduced with data sparsity.

**Fig. 3** Research gaps and solutions of groundwater conservation

how blockchain, ML, and DL have changed the very definition of groundwater conservation. In particular, blockchain ensured data integrity and its management, while ML predicted and mapped groundwater quality and delivered results through remote sensing. DL improved GLP, recharge estimation, and anomaly detection. Important benefits for sustainable GWM were highlighted, while there remained further tasks on concrete specification and developing their applications.

Wastewater Treatment

This analyzes WWT and looks into the various uses of blockchain technology in this field. It discusses topics like the recycling and reusing of WWT processes and leveraging ML for wastewater quality assessment. Besides, it covers the forecasting and monitoring of wastewater quality through DL techniques. The report also examines the DL aids in detecting fouling in membrane bioreactors.

Table 7 Comparison of wastewater treatment, recycling, and reuse in blockchain

Refs	Data collection	Pre-processing	Execution time	Computational efficiency	Limitations
(Tiwari et al. 2020)	Water management system data.	✓	Fast	High	Had limited scalability for large-scale water networks.
(Bamakan et al. 2022)	Data on water sustainability and circular economy barriers.	✗	Medium	Low	Faced complex barriers in adopting blockchain in traditional water systems.
(Senthil Rathi and Senthil Kumar 2022)	Hospital waste management data.	✓	Medium	High	Was dependent on hospital infrastructure and blockchain adoption rate.
(Liu and Shang 2022)	WWT data from the textile industry.	✗	Slow	High	Was limited by specific industry focus, leading to scalability challenges.
(Pandey et al. 2022)	Agricultural water trade data.	✓	Fast	High	Required extensive setup for blockchain infrastructure in agricultural regions.
(Jiang et al. 2023)	WWT data for smart cities.	✓	Medium	Low	Encountered implementation challenges in non-smart city regions.
(Sosnowski and Sepczuk 2023)	Waste management and recycling data.	✓	Fast	High	Incurred high initial setup costs for blockchain systems.
(Lin et al. 2024)	Recycling data for the reuse process.	✓	Fast	High	It was limited by the failover system complexity and the blockchain adoption rate.

Blockchain Applications in Wastewater Treatment, Recycling, and Reuse Treatment

Blockchain technology in water and waste management has become an important topic of study in recent times, with such technology helping increase efficiency and transparency while still being sustainable. Tiwari et al. (2020) developed a framework for smart contracts aimed at decentralized water management systems. Bamakan et al. (2022) extended blockchain applications to hospital waste management, proposing a system that ensures safe and efficient waste disposal while addressing future trends in waste management practices. Senthil Rathi and Senthil Kumar (2022) examined the future of blockchain in WWT within the textile industry, focusing on how this technology improves environmental sustainability by enhancing the traceability and accountability of WWT processes.

Liu and Shang (2022) applied blockchain technology to agricultural water rights trade management, demonstrating its capacity to streamline and secure the trade of water rights, which was crucial for sustainable agricultural practices. Pandey et al. (2022) discussed WWT with a focus on smart cities, exploring the technical interventions, including blockchain, that optimize WWM systems in more sustainable urban environments. Jiang et al. (2023) discussed the role of blockchain in waste management in detail, pointing out the challenges limiting its adoption and opportunities that this technology presents, particularly in terms of improving waste tracking, reducing fraud, and enhancing compliance with environmental regulations. Sosnowski and Sepczuk (2023) developed Smart User-centered Recycling and Reuse Environment (SURE), a blockchain-based solution for secure recycling and reuse. Lin et al. (2024) created

a blockchain system to manage construction waste management. Table 7 illustrates the comparison of WWT, recycling, and reuse treatment in blockchain applications.

In Table 7, pre-processing refers to the steps to prepare and clean the data before applying ML models, like handling missing values or normalizing data to enhance model performance. When marked with ✓, it indicates that pre-processing techniques were applied, which ✗ means that no pre-processing was utilized.

DL-based Wastewater Quality Prediction and Monitoring

The reviewed studies contributed to advancing the prediction and monitoring of WWT and water quality using DL. Jafar et al. (2022) utilized shallow and deep ANN models for predicting effluent quality in full-scale wastewater treatment facilities. Cao et al. (2023) developed a DL-based Soft Sensor (DLSS) to forecast microbial treatment efficiency. Yang et al. (2023) applied DL to the prediction of Effluent Quality in a Constructed Wetland (EQCW). Jadhav et al. (2023) explored current trends and challenges in using ANN for water and WQP (ANN-WQP). Shaban et al. (2024) implemented advanced DL networks for Real-time WQP (RWQPP). Xie et al. (2024) suggested a hybrid DL approach to enhance Real-Time Effluent Quality Prediction (RTEQP). Mokarram et al. (2024) integrated DL and Fuzzy method Integration for Water Quality Monitoring (FIWQM) to improve water quality monitoring. Table 8 explains the comparison of wastewater quality prediction and monitoring.

In Table 8, ✓ in the data collection column signifies that data collection techniques were used. Preprocessing refers

Table 8 Comparison of wastewater quality prediction and monitoring

Refs	DL technique	Data Collection	Pre-processing	Performances	Limitations
(Jafar et al. 2022)	Shallow and deep ANN	✓	✓	Achieved prediction accuracy for effluent quality and demonstrated model robustness.	Limited generalizability to smaller plants; computational complexity increased with deeper networks.
(Cao et al. 2023)	DL-based Soft Sensor	✓	✓	Demonstrated efficiency in prediction accuracy and time-series performance.	Required significant computational resources; sensitive to noisy data.
(Yang et al. 2023)	DL for Constructed Wetland Effluent Quality Prediction	✓	✓	Showed high effluent quality prediction accuracy and model sensitivity.	Limited to constructed wetlands; required high-quality real-time data.
(Jadhav et al. 2023)	ANN	✓	✓	Maintained prediction accuracy and consistency across different systems.	Faced challenges with real-time prediction; performance highly dependent on data quality.
(Shaban et al. 2024)	Advanced DL Networks	✓	✓	Achieved real-time prediction accuracy with low latency.	Limited by data availability; required extensive training time.
(Xie et al. 2024)	Hybrid DL Approach	✓	✓	Demonstrated effluent quality prediction accuracy and real-time adaptability.	Required manual feature engineering; complex architecture increased computational load.
(Mokarram et al. 2024)	DL Neural Networks combined with Fuzzy Method	✓	✓	Enhanced water quality monitoring accuracy and improved model adaptability.	Complexity of the fuzzy method integration; high computational requirements.

to techniques used to adjust model parameters and enhance overall predictive accuracy. A ✓ in the preprocessing column indicates that preprocessing methods were applied.

DL-based Fouling Detection in Membrane Bioreactors

The reviewed studies focused on advancements in membrane fouling prediction and management in membrane bioreactor systems. Hamed et al. (2019) applied new deterministic methods of systematic investigation for fouling in membrane bioreactors, bringing in unique approaches in analyzing fouling incidents. Park et al. (2019) went for DNN modeling in estimating the increase in fouling and associated flux loss in nanofiltration and reverse osmosis membrane systems. Viet et al. (2021) created a DL-based approach combining an osmotic membrane bioreactor with a clarifier to effectively manage salt accumulation and membrane fouling. Wang et al. (2023) suggested an improved residual neural network for the diagnosis of membrane fouling in membrane bioreactors, thus increasing diagnostic accuracy. Wang and Li (2024) utilized an ANN for forecasting fouling behavior in large anaerobic membrane bioreactors processing municipal wastewater. Table 9 illustrates the comparison of fouling detection in membrane bioreactors.

Figure 4 highlights the research gaps and solutions of WWT. Real-world examples provide an opportunity to understand their practical application toward the proposed solution. For example, a smart city project that connects

IoT sensors to blockchain results in better data integration and transparency by enabling real-time tracking of variables such as energy consumption or traffic flow while providing tamper-proof data management with blockchain. Hybrid models for anomaly detection include live monitoring and unexpected event detection in industrial setups.

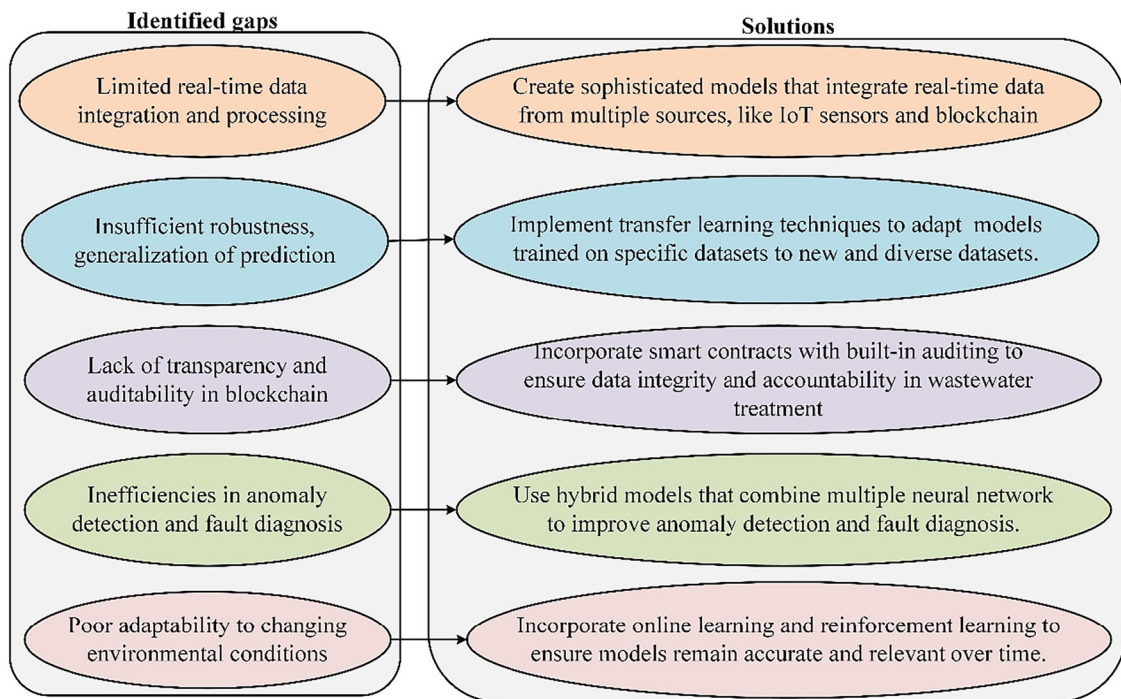
Additionally, online reinforcement learning for adaptive traffic management was a showcase of the online, dynamic adjustment of a system of traffic signals, reflecting actual traffic conditions, further validating the utility and practicality of the solutions proposed.

Groundwater Conservation Dataset-based Review

Technological advances in GWM are providing better data resolution and integration of systems that result in improved management and efficiency. Ali et al. (2021), for instance, improved the resolution of Gravity Recovery and Climate Experiment (GRCE) data to assess groundwater storage that borrowing from data, accurate estimates of groundwater depletion that were not possible before due to remote sensing. Abd El-Aziz et al. (2021) included an intelligent recommender system for GWM that used decision support tools to ensure that resources were allocated efficiently. Vishwakarma et al. (2021) used GIS-based multicriteria evaluation to enhance site selection strategies by prioritizing MAR sites in the semi-arid regions of India. Yin et al. (2022) used ML algorithms and hydrological systems to enhance GRCE-based estimates of groundwater storage by

Table 9 Comparison of fouling detection in membrane bioreactors

Refs	DL techniques used	Data Collection	Performance	Limitations
(Hamed et al. 2019)	Used deterministic models for fouling investigation.	Collected data from membrane bioreactors through experimental studies in chemical engineering processes.	Accuracy of fouling detection, membrane performance degradation.	Focused on deterministic models, not DL approaches; applicability limited to specific bioreactor configurations.
(Park et al. 2019)	Applied a DNN for fouling and flux decline.	Obtained data from NF/RO membrane filtration systems during real-time filtration.	Flux decline prediction, fouling growth accuracy.	Relies on abundant labeled data and intensive computation for training.
(Viet et al. 2021)	Integrated DL with an osmotic membrane bioreactor–clarifier system.	Gathered data from osmotic membrane bioreactor experiments with various salt concentrations.	Salt accumulation reduction, membrane fouling reduction.	Depended on complex experimental setups; limited generalizability to other systems.
(Wang et al. 2023)	Developed an improved ResNet for fouling diagnosis.	Recorded real-time membrane bioreactor data.	Fouling prediction accuracy and diagnosis speed.	Required substantial data preprocessing, not suitable for small datasets.
(Wang and Li 2024)	Employed ANN for fouling prediction in anaerobic conditions.	Analyzed pilot-scale data from municipal WWT plants.	Predictive accuracy for fouling, real-time performance.	Required large-scale pilot testing, limited scalability for smaller systems.

**Fig. 4** Research gaps and solutions of wastewater treatment

mitigating spatial resolution issues. Sarkar (2023) used GIS - Multi-Criteria Decision Analysis (GIS-MCDA) to evaluate regions facing water stress in the Godavari basin. Cosgrove et al. (2024) advanced operational hydrology with the National Oceanic and Atmospheric Administration (NOAA) National Water Model (NWM), which improved continental-scale water resource modeling. Badetiya and Barale (2024) focus on modeling groundwater levels using Geographically Weighted Regression (GWR). Table 10

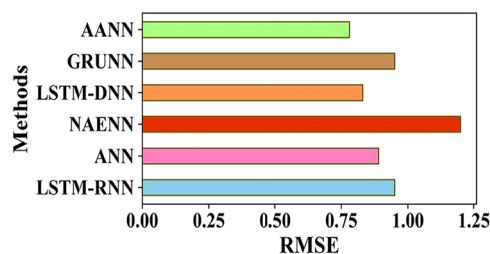
illustrates the comparison of groundwater conservation with datasets.

Results and Discussions

This section presents a comparison of the various techniques like SCBWMA, AI-BWM, CEIB-PAS, DCEWS, BUWMD, AANN, GRUNN, LSTM-DNN, NAENN, LSTM-RNN,

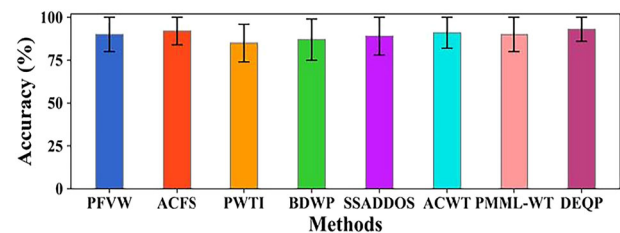
Table 10 Comparison of groundwater conservation with datasets

Refs	Techniques used	Datasets Used	Performances	Limitations
(Ali et al. 2021)	Improved GRCE data resolution.	GRCE dataset.	Improved spatial resolution.	Required accurate hydrological models for best results.
(Abd El-Aziz et al. 2021)	Developed an intelligent GWM recommender system.	Indian groundwater data.	Enhanced accuracy and recommender system performance.	It was limited to specific regional datasets.
(Vishwakarma et al. 2021)	Applied a geographic information system-based multicriteria evaluation strategy.	Geographic information system data, semi-arid environment data.	Increased site prioritization efficiency.	Lacked consideration for all local hydrogeological variations.
(Yin et al. 2022)	Integrated ML and a hydrological model.	GRCE data, integrated with hydrological model information	Enhanced spatial resolution and accuracy of storage estimates.	It was dependent on the quality of hydrological model inputs.
(Sarkar 2023)	Adopted a decision analysis method grounded in GIS and multi-criteria assessment.	Water stress data, Godavari basin data.	Appraised water stress regions.	The lack of extensive regional data affected accuracy.
(Cosgrove et al. 2024)	Employed NOAA data.	National water dataset.	Improved hydrological modeling performance.	It was computationally intensive and needed extensive resources.
(Badetiya and Barale 2024)	Implemented geographically weighted regression.	Groundwater level data.	Achieved regression accuracy.	Lacked consideration for spatial variability in data.

**Fig. 5** RMSE analysis of GLP using neural networks

PFVW, ACFS, PWTI, BDWP, SSADDOS, ACWT, PMML-WT, DEQP, DLSS, EQCW, ANN-WQP, RWQPP, RTEQP, and FIWQM. Selection of these methods was guided by their applicability to GWM and their effectiveness in water quality forecasting and surveillance, and their effectiveness in utilizing ML and DL methodologies. While these models demonstrate varied performance, to understand their real-world utility, one examines the factors that determine their success, such as data characteristics, model complexity, and training methods. This evaluation helps identify best practices and inherent limitations within each approach.

Figure 5 presents an analysis of the RMSE for neural network-based GLP models. Among these, AANN demonstrates superior performance, showcasing its potential for highly accurate predictions. Other models, like GRUNN and LSTM-DNN, perform well but with slightly higher errors. In contrast, models like NAENN present greater challenges in prediction accuracy. This suggests that although advanced neural architectures generally improve prediction, effectiveness depends on how well models capture temporal dependencies and the quality of training

**Fig. 6** Accuracy analysis of ML-based predictive modeling of wastewater

data. NAENN's poorer performance results from overfitting or inadequate feature representation, indicating a need for further optimization. The overall trend suggests that neural network models significantly improve GLP, with AANN emerging as the most reliable option. However, these results require careful interpretation due to possible dataset limitations and inconsistent validation practices across studies, which affect the generalizability of the findings.

Assessment of various ML techniques to predict wastewater quality through accuracy metrics in Fig. 6. DEQP leads the comparison, with high accuracy suggesting its effectiveness for prediction in wastewater systems. Methods like ACFS and ACWT follow closely, while models like PWTI show lower performance. The strong performance of DEQP likely stems from its ability to handle nonlinearities inherent in wastewater parameters. Conversely, the lower accuracy of PWTI indicates potential limitations in managing complex or noisy data, highlighting the need for enhanced features.

Figure 7 compares the performance of DL-based methods in wastewater quality prediction, emphasizing the

superior performance of RLTEP. RLTEP outperforms other techniques by minimizing both RMSE and MAE, making it the most promising DL model for accurate water quality monitoring. While other models, such as DLSS and ANN-WQP, exhibit higher error rates, these findings demonstrate DL's significant potential in improving prediction precision. However, higher error rates in some DL models highlight challenges related to model optimization and possible overfitting, especially under limited training data conditions.

The performance of multiple blockchain applications is assessed in Fig. 8 by examining precision, recall, accuracy, and F1-score. DCEWS has excelled in all measures and is the most effective method of implementing blockchain applications for water quality management. The utilization

of blockchain technology by DCEWS introduces more transparency and efficiency in monitoring and regulatory processes through sustainable and resilient GWM and wastewater systems. Despite these advantages, blockchain implementations such as DCEWS face challenges including computational overhead, data privacy concerns, and integration complexity with existing infrastructure. Overcoming these obstacles is essential for widespread adoption.

Overall, while the reviewed techniques show promising advances and model capabilities, several limitations persist. Variability in dataset size, quality, and regional factors influences model robustness. Many studies lack comprehensive cross-validation or real-time deployment evaluation, which limits assessment of practical utility. Future research focuses on improving model generalizability, integrating ML and DL methods, and addressing deployment challenges to enhance GWM and WWM systems effectively. Also, will explore hybrid or ensemble approaches to capitalize on the complementary strengths of different architectures.

Bibliometric Analysis

The bibliometric analysis offers a structured summary of recent scholarly contributions intensive on GWM and WWM using advanced technologies such as blockchain and DL. It highlights key publication sources, author

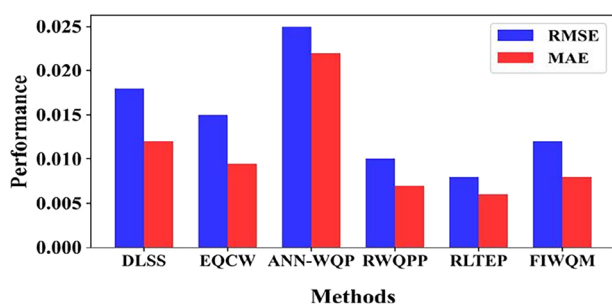
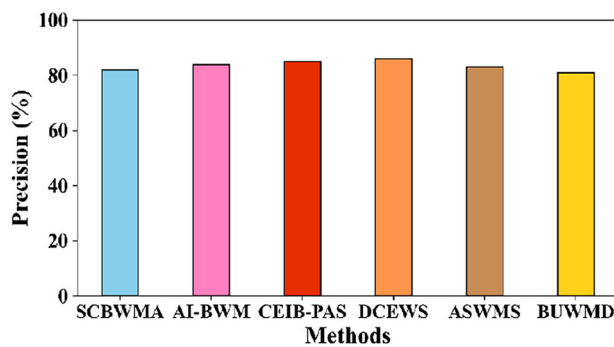
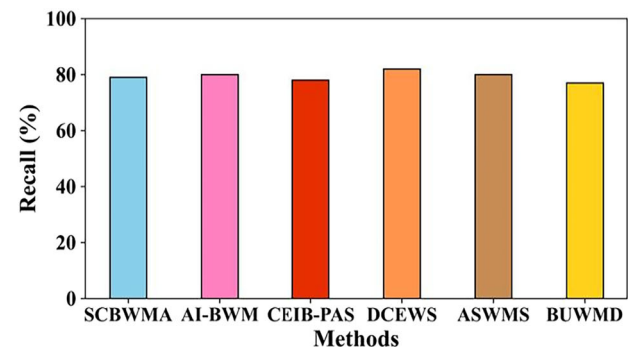


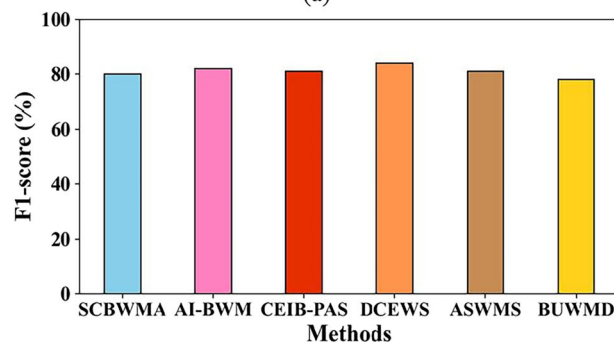
Fig. 7 Comparison of DL-based wastewater quality prediction and monitoring



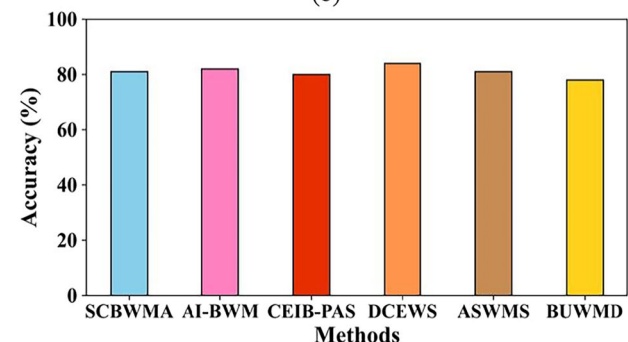
(a)



(b)



(c)



(d)

Fig. 8 Comparison of **a** precision, **b** recall, **c** F1-score, and **d** accuracy in blockchain applications

Table 11 Bibliometric analysis of blockchain and DL applications in WWM and GWM

Authors	Journal	Year	Citation count	Application area
Sundaresan S, Suresh Kumar K, Ananth Kumar T, Ashok V, Golden Julie E	Blockchain Smart Cities	2021	21	Intelligent water management in smart cities using blockchain technology
Di Nunno F, Granata F	Environmental Research	2020	143	GLP using neural network in the Apulia region
Chen H-Y, Vojinovic Z, Lo W, Lee J-W	Water	2023	18	GLP using DL methods
Naqash M, Syed T, Alqahtani S, Siddiqui M, Alzahrani A, Nauman M	Urban Science	2023	19	Efficient urban water control and leak tracking via blockchain
Liu Q, Yang L, Yang M	Sustainability	2021	44	Digital innovations and circular economy barriers in sustainable water systems
Shaban WM, Xie D, Elbaz K, Shen S-L	Journal of Water Process Engineering	2024	3	AI-enabled forecasting of wastewater quality in real-time
Xie Y, Chen Y, Wei Q, Yin H	Water Research	2024	43	Real-time effluent prediction in WWTPs via hybrid DL

contributions, application domains, and the relevance of each review within the evolving landscape of smart water systems.

Table 11 presents a bibliometric overview of some of the key studies applying blockchain and DL technologies in various WWM domains, including GLP and wastewater monitoring. It summarizes the authors, publication sources, citation counts, and specific application areas, highlighting the growing academic focus on intelligent and sustainable water solutions.

Practical and Theoretical Implications

The results suggest important practical and conceptual value. Especially, models like AANN and RLTEP demonstrate strong predictive accuracy, indicating their suitability for deployment in real-time groundwater level and wastewater quality monitoring systems. These models support proactive decision-making for water resource managers by facilitating the prompt identification of irregularities and patterns. The integration of blockchain frameworks, particularly DCEWS, introduces secure, transparent, and decentralized data handling, which enhances trust and regulatory compliance in water quality management. Theoretically, the performance variation across models underscores the need to consider model–data compatibility, architectural complexity, and optimization strategies. The observed superiority of specific neural and DL architectures emphasizes the growing relevance of data-driven approaches over traditional hydrological models. Moreover, this review reinforces the value of hybridizing ML and DL with domain-specific knowledge to address the multifaceted challenges of groundwater conservation and WWT. These insights guide future research towards more adaptive, explainable,

and scalable models tailored for sustainable WWM systems.

Conclusion

This review has examined various applications of blockchain, ML, and DL technologies for groundwater conservation and WWM, emphasizing innovations from 2019 to 2025. A total of 97 papers were reviewed for this review, based on the technologies in sustainable water resource management. Blockchain technology increased the security, transparency, and regulatory adherence of the water system. Due to the noncentralized nature of blockchain, immutable records have been guaranteed, thus ensuring trust for groundwater observation and WWT.

Processes such as wastewater reuse and recycling benefit from blockchain's influence, promoting efficiency and compliance. Additionally, advancements in predictive modeling, anomaly detection, and resource optimization have been driven by ML and DL. Applications include GQP, resource mapping, and optimization of water distribution networks, highlighting novel approaches for real-time monitoring and resource allocation. However, this review identifies certain limitations, including the scalability challenges of blockchain implementations, the computational demands of ML/DL models, and gaps in real-world deployment and integration. Future research focuses on integrating smart contracts to automate regulatory compliance and improve operational efficiency. Further exploration of real-time surveillance systems integrated with IoT-based governance frameworks will be crucial to enhance responsiveness and utilize data to guide decisions in managing water resources. Addressing advancements in efficient and long-term GWM and WWM solutions.

Data Availability

No datasets were generated or analysed during the current study.

Author contributions Amit Mithal: Conceptualization, Methodology, Validation Blessy Thankachan: Project administration, Supervision, Investigation

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Compliance with Ethical Standards

Conflict of Interest The authors declare no competing interests.

Ethical Approval All applicable institutional and/or national guidelines for the care and use of animals were followed.

Informed Consent For this type of analysis formal consent is not needed.

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